How Platform Pre-Announcements Reshape Complementor Competition: Evidence from Google's In-App Purchase Regulation

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Abstract

In 2024, a U.S. federal court ordered Google to expand payment options for Android apps, following controversy over its 2020 announcement mandating Google's billing system. While legal disputes continue, there is limited empirical evidence on how such pre-announcement and policy changes affect platform markets. Using large-scale mobile app panel data from Android-dominant regions , this study examines the impact of Google's pre-announcement and subsequent enforcement on complementors' new user acquisition. We find that even prior to enforcement, the announcement led new users to favor apps aligned with Google's regulation, reinforcing Google's market power. In particular, well-established apps and those with diversified revenue streams gained more new users. This effect was especially pronounced among high-purchasing-power users, who tended to select compliant and established apps. These findings highlight that policy pre-announcements can significantly influence competitive dynamics, providing novel empirical insights into how platform owners' strategic signals reshape market outcomes and implications for platform governance.

Keywords: Platform Governance, Platform Policy Intervention, In-App Purchase, Pre-Announcement Effects, Ex-Ante Responses, Digital Platforms, Mobile App Ecosystem

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1. Introduction

Google continues to face significant legal challenges in U.S. courts. In July 2024, the federal court in Washington, D.C., ruled in favor of the U.S. Department of Justice in its antitrust lawsuit against Google over its dominance in the search engine market, stating, “Google is a monopolist, and it has acted as one to maintain its monopoly” [44]. While this case focuses on Google’s dominance in the search engine market, the company has continued to strengthen its control over the broader digital ecosystem. In particular, its in-app purchase system has also become a focal point in antitrust debates. Amid this ongoing scrutiny, in October 2024, a separate federal court in San Francisco ruled that Google must allow external payment options (i.e., third-party payment options) for Android users in the app store. In response, Google announced plans to appeal the decision, and the legal battle continues [7].

The controversy over Google’s strengthened in-app purchase policy (hereafter the IAPP), which is central to this ruling, dates back to July 2020. At that time, Google announced a strict regulation requiring app developers selling digital goods and services on its platform to mandatorily use its proprietary billing system. Previously, Google recommended using its billing system for all in-app purchases, but this was a suggestion without concrete penalties or enforcement measures for non-compliance. As a result, while some apps followed Google’s recommendation, some developers opted for third-party payment systems (e.g., Stripe or PayPal), benefiting from lower transaction fees and avoiding Google’s revenue monitoring. However, the situation changed drastically when Google announced in July 2020 that it would no longer allow third-party payment systems. This policy effectively restricted app consumers from choosing their preferred payment methods when purchasing digital content, such as streaming services, paid apps, and add-ons. From that point forward, all in-app purchases would need to be processed exclusively through

Google's billing system starting from an expected enforcement date[1], allowing the platform owner to impose a 30% commission, often referred to as a "toll." Additionally, Google made it clear that strict enforcement measures would be imposed on developers who did not comply with the new policy. Non-compliant apps faced restrictions on distributing updates or being listed on the Google Play Store (GPS), and in the worst-case scenario, they could be entirely removed from the app marketplace [20].

In defense of this strategy, the technology giant avers that enforcement is inevitable because of the rising cost structures associated with the operation and governance of the expanded ecosystem and resources within the Android platform [52]. If we subscribe to this explanation, then the additional revenues earned from the payment mandate, as it turns out, facilitate sales generation on developer platforms. Notwithstanding these claims, the plan for strict implementation of the new mobile payment structure has triggered a considerable public backlash and regulatory scrutiny. Government agencies and regulators all around the world view the new payment requirement as constituting an exploitation of a dominant market position by means of practice that is anti-competition and have accordingly established legislation to curb control by major app store operators [69].

When the plan becomes a reality, however, app developers have few feasible options available other than to comply with Google's policy. They can be kicked out from the only app market with no alternatives unless they replace the 3rd parties billing system with those from the platform owner. Even if they barely avoid the risk of being removed from the Google Play Store by adopting Google's billing system, those now subject to a much higher "revenue-sharing" with the platform owner will inevitably increase prices and fees. Then, increased costs may inevitably be passed on to billions of consumers around the globe who

[1] In its initial official announcement, Google required developers to use its own billing system for in-app purchases by September 30, 2021 [6].

have no alternatives for securing app services. For instance, the new payment intervention is expected to drive up the prices of over-the-top (OTT) subscriptions, music streaming services, and other content-driven transactions. In the long run, the new policy may alter developers' business models and competition dynamics. Amid such restructuring, many small, independent app developers, who provide valuable resources under numerous content categories to the Google platform, are concerned that their battles with platform owners will end in defeat, given the imbalance in power.

Despite the importance and business implications of the new policy initiation, the literature has been reticent on these matters partly because the enforcement of platform-driven in-app purchase is a new phenomenon in flux, that is, in a state of ongoing progression and adjustment.[2] Our understanding is still rudimentary with respect to how such major interventions can disrupt the market equilibrium in the mobile app industry and cause shocks to the welfare of platform participants. The present study is therefore motivated by the need for a more nuanced investigation of the effects of Google's payment policy on platform stakeholders, shedding valuable light on policy debates surrounding platform owners' leveraging of their dominance in the market [47, 66].

Furthermore, whereas prior studies have primarily focused exclusively on the reactions of complementors (i.e., app developers) to the possible actions of platform owners in unraveling the dynamic interplay between these stakeholders (e.g., [12, 49, 72]), the current research shifts attention to the demand side's reaction. Accordingly, we demonstrate that Google's pre-announcement, which may initially appear to serve merely as a cautionary notice to complementors, can function as a strategic mechanism that induces a preemptive

[2] Because Apple strictly implemented the in-app payment requirement from the initiation of service provision, developers and users took the fees charged by the company for granted. By contrast, such rules existed only in name at the Google Play Store (GPS), with Google implementing it in a more relaxed manner. As a result, the announcement of the strict enforcement has come under fire from developers, consumers, and regulators.

shift in demand, thereby exerting pressure on complementors. This perspective underscores that pre-announcements are not merely disclosures of impending policy changes but can actively reshape market competition by altering user behavior in ways that influence complementors' strategic positioning.

To theoretically assess the aforementioned issues, we draw on previous works that pertain to the governance and the strategic actions of digital platform owners in their pursuit of expanded market power (e.g., [11, 25, 66]). For instance, Wen and Zhu [66] investigate how platform developers react to Google's *threat of entry* into their marketplaces through the company's offering of similar products. The authors demonstrate that platform complementors respond to the potential move of platform owners from the onset of policy intervention (i.e., when the threat of policy enforcement increases but its actual implementation has not yet occurred) rather than wait until its actual implementation. This tendency is attributed to the fact that acting early more effectively smooths out the settlement of expenses incurred from transitioning over time and enables developers to gain advantages over their competitors [71]. This line of research is extended in our study by delving into how the threats of payment enforcement affect the *ex-ante* reaction of the demand side, which also can change the competition relationship between complementors, addressing the multi-sided characteristics of platform economics.

For empirical validations, we acquired large-scale panel data on smartphone app usage from Nielson KoreanClick, a global branch of the Nielson Company, which monitored and measured activity on mobile apps installed on consumers' smartphones [21]. An investigation involving Korean mobile consumers is ideal for a number of reasons. Given that the Android platform accounts for over 70% of the market share in Korea when it comes to mobile operating systems [59], the effects of Google's IAPP on platform ecosystems and participants are substantial. The country also has numerous mobile app developers that substantially vary

in size, domestic platforms (i.e., incumbent developers), and an app market platform (i.e., ONE Store) positioned as an alternative to the GPS, albeit this platform is still in an embryonic stage. These characteristics enable us to comprehensively assess the possible effects of Google's strategic extension of its dominance in app markets through the imposition of the in-app purchase structure.

We observed the impacts of the *pre-announcement* of IAPP on the acquisition of new users, which is an element crucial to the growth and innovativeness of platforms [5]. The key findings reveal that the pre-announcement of the in-app purchase enforcement increases the number of new users acquired, particularly for apps that have already adopted Google's billing system. This shows that only the pre-announcement on planned enforcement can bring more favorable opportunities for apps that comply with Google's control, implying platform owners' imbalance of power even without implementation. We also identified the heterogeneous effects among developers in accordance with their popularity and revenue models. Google's pre-announcement attracted more new users for apps ranked popular based on their user base, thereby intensifying polarization contingent on the popularity of apps.

Additionally, in terms of revenue models, the Google announcement unfavorably influences developers who rely on paid apps as a single source of revenue. Interestingly, however, the announcement results in a sizable increase in the number of new users among developers containing in-app purchase apps, who profit from both install fees and advertisements. Alongside, we found that the pre-announcement brought new users with relatively *high purchasing power* and *strong involvement* in the mobile app usage of the IAP apps. Furthermore, this inclination is strengthened when IAP apps are ranked more popular. Consequently, more established IAP apps that have a strong user base acquired more new users in both larger quantity and better quality after the pre-announcement of enforcement from the platform owner. Lastly, we also analyzed the impacts of IAPP *implementation* on

new user acquisition as well, leveraging the fact that Google terminated the grace periods and initiated the penalty for apps using 3rd party billing system in April 2022. From the results, we observe that the direction of demand spillover after the implementation is consistent with the results after the pre-announcement, implying the pre-announcement has the potential power to lead strategic reshaping in demand without direct implementation. On the basis of these findings, we provide platform owners, regulators, and developers with useful implications regarding how payment-related intervention can affect the dynamics of mobile-based platform ecosystems and the welfare of their participants.

Our study offers several contributions to the growing literature on digital platform governance. First, Google's recent regulation mandating in-app purchases is a paradigm shift that significantly impacts all participants in this two-sided platform. Despite the importance of such policy interventions for global mobile ecosystems, there is limited understanding of how platform-initiated policies affect complementors (i.e., developers). Our research sheds light on *how* Google's mandatory payment structure influences complementors and *who* benefits and who suffers from these interventions, along with the actions needed to support small developers, consumers, and the overall health of the platform ecosystem. Using comprehensive mobile app data from Nielsen, we add initial insights into these pressing issues, which are of great interest to digital and mobile platform stakeholders. Secondly, while most platform studies have focused on complementors' subsequent behavior, such as innovation or market strategy, impacts on the demand spillover between complementors (i.e., new user acquisition) are relatively underexplored. We address this gap by examining the influence of the payment mandate on developers, with particular attention to how the policy affects compliers versus non-compliers and incumbents versus newer developers. We find that the in-app purchase regulation may ultimately benefit developers favored by platform owners or those already established, while hindering newer developers who face pressure

from both platform owners and incumbent developers. Lastly, our study broadens the scope of pre-announcement effects within platform governance. We examine how platform owners can shift user demand via pre-announcement even before implementation, underscoring the importance of monitoring these announcements, as they alone can alter market competition in the platform's favor. In summary, our study provides valuable implications for policymakers, platform owners, developers, and consumers regarding the impact of mandatory in-app purchase requirements on platform ecosystems, stakeholder well-being, and sustainable platform growth.

2. Theoretical Background and Related Literature

2.1. Platform Governance

Research on platform governance abounds with numerous conceptual prototypes and empirical manifestations (e.g., [17, 51, 53, 61]). In these works, the central issue is how platforms' orchestration and control of complementary *extensions*[3] determine the growth and success of their ecosystems [61, 68]. Following previous studies, we define a platform as "a system that brings system adopters together with firms that provide complements to the system, so-called complementors" [17, p.445]. As a platform operates on the basis of modularity-driven architecture[4] and multi-sided interaction, a collection of complementary resources, products, and services available within and across its ecosystem augments its current value and functionality, as well as shapes its future innovation and evolution [8].

To begin with, digital platforms, such as the GPS, offer a technological foundation or interface (i.e., the Android operating system) by which complementors (i.e., app developers) create and enhance numerous generative extensions via IT artifacts (i.e., mobile apps) [70].

[3] A complementary extension is any product or service that "extends" the value and attractiveness of a focal platform [6, 61].
[4] Modularity refers to the degree to which a product or system can be deposed into modules (subsystems) that can be readily recombined as needed [54], and this concept has served as a foundational mechanism for the construction and operation of a platform ecosystem [3].

For example, the GPS oversees the development and maintenance of the Android interface, through which thousands of complementors use platform resources to develop apps falling under numerous categories. Additionally, such platforms serve as intermediaries that facilitate the commercial exchange of digital goods between complementors and end users [19, 22] while collecting feedback and new ideas from the latter in pursuit of extension evolution (i.e., upgraded versions) and other innovations.

Over the years, researchers have carried out a multitude of investigations on platform governance, which can be broadly categorized into two streams that are particularly germane to our work: the infrastructural design and engineering of a platform and platform owners' management of complementors.

2.1.1. Platform Design and Complementor Relationships

One of the core issues that has been intensively debated among researchers of platform architectures is *how open* a platform should be when it comes to granting access and devolving control to complementors [5, 13, 31, 47, 62] . Choosing the "right" degree of openness may entail a trade-off between adoption and appropriation [67]—a choice that becomes even more challenging given the evolutionary nature of a platform. Earlier works in favor of an open policy in platform operation demonstrated that granting extensive access to complementors that interoperate can foster vibrant environments in which fresh and diverse ideas are born and active experimentations are encouraged (e.g., [3, 15, 16, 64]). Pursuing an open strategy, however, comes with a cost because it can shrink an inventor's share of profits by reducing entry barriers and intensifying intra-platform competition [67]. This argument is supported by Boudreau [5], who finds an inverted U-shaped relationship between innovation and the level of access accorded to complementors; that is, reasonable openness in an interface or a market increases developers' incentive to innovate, but too much backfires and eats away at profits. This non-monotonic relationship is attributed to the transaction and

coordination costs between a platform owner and complementors [5].

As a means of overcoming entry barriers in open platforms with strong network effects and high switching costs, a "platform envelopment" strategy is presented by Eisenmann et al. [13] for situations wherein firms can leverage their existing user bases to enter another market that requires a critical mass of users. With platform envelopment, entrants can successfully penetrate an open market and leapfrog over incumbents, even in the presence of strong network effects. Using game-theoretic stylized facts, Parker and Van Alstyne [47] develop a sequential innovation model that reflects what the *openness* of a platform economically means to platform owners and developers. More specifically, the findings show that closing a platform enables platform owners to increase their eligibility to charge for access, whereas opening it empowers platform developers to leverage a larger pool of resources. The argument for openness is even more reinforced by the fact that the longer developers retain legal rights and protections over their innovations, the higher the economic gains (i.e., royalties) earned by both parties.

The in-app purchase system initiated by Google could pose significant ramifications for the design of the Android-enabled platform, as it may radically shift the equilibrium between openness and closure toward the latter end of this continuum. It may place many independent, non-compliant, or small developers at an additional risk given the increased commission payment to the owner and the threat of being removed from the Google Play Store if they do not comply. The in-app purchase restriction may consequently increase barriers to new entrants, which can reduce diversity in the resource pool and soften competition. As a platform diminishes in size and diversity, the interface on which complementors collaborate for common benefits erodes in function and value creation [50]. Contrastingly, the platform closure driven by the in-app purchase requirement potentially gives rise to new opportunities for incumbent complementors. Established players

may leverage the platform envelopment strategy (i.e., putting an existing user base to advantage for market entry) [13] as they penetrate the market that their small counterparts are compelled to leave. Moreover, the policy intervention enables incumbents to aggressively acquire small developers with high innovation capabilities at relatively lower prices, thereby strengthening their bargaining position against a platform owner.

2.1.2. Platform Owners' Actions and Responses

Researchers have also directed their attention to platform owners' management and control over complementors. This stream of scholarship is particularly relevant to the current study, given that the in-app purchase requirement can be viewed as a strategic move by a platform owner to control complementors. These studies generally focus on *platform owners' actions* implemented within the ecosystem that immediately alter the operating environment for complementors. Specifically, Malgonde et al. [39] highlighted that algorithmic decisions by platform owners, such as recommendation systems, can create imbalanced outcomes across stakeholders, reinforcing the idea that seemingly neutral platform actions may have uneven impacts on complementors. Furthermore, Li and Agarwal [36] explored the outcomes of Facebook's integration of its own photo-sharing app, Instagram (i.e., first-party applications), and discovered an asymmetric pattern of demand spillover. The integration increases the number of active Instagram users, which, in turn, has a positive spillover effect on large third-party apps but a negative spillover effect on small third-party apps in Facebook's ecosystem. Zhu and Liu [72] assessed how Amazon's entry into complementary markets affects the latter's ex-post business strategies. Amazon selectively targets successful product spaces and penetrates only those that entail less effort to grow, demonstrating that the largest online commerce platform in the world leverages complementors' platform-specific investments in making entry decisions. Foerderer et al. [17] examined how platform entry affects innovation efforts, finding that Google's entry into the photo app market through Android GPS in 2015

led to a nearly 10% increase in major updates for affected apps.

As mentioned, while many studies have focused on the implemented action of platform owners, limited research has explored the effects of *anticipated interventions* or *threats*–scenarios where a platform owner signals a change but has not yet implemented it. However, unlike market entry or integration, which physically reconfigures competition, pre-announcements about scheduled or anticipated interventions from platform owners operate on the level of expectation, potentially altering the market dynamics even before any tangible changes occur. In this regard, our study contributes to expanding the literature on platform governance by exploring platform owners' announced, but not yet implemented anticipatory interventions that have not been explored much.

2.2. *Ex-Ante* Responses to Threats

Our study aligns closely with research on forward-looking behaviors and *ex-ante* strategic responses to specific events. Studies in this area have primarily explored how users and firms proactively adjust their behaviors and strategies in response to anticipated environmental changes. However, most of this research has focused on firms' ex-ante responses to threats rather than individual consumers' ex-ante responses. Given that responsive strategic formulation and implementation cannot be effectively established in short periods, companies are supposed to respond to expected changes at the early stage of environmental variation—adjusting their strategies and preparing for these upcoming variations—to cope with these developments, overcome risk before an actual change is implemented, and achieve a competitive advantage over late responders [71]. Examples of research investigating firms' ex-ante response to external threats include studies by Goolsbee and Syverson [21], Pan et al. [46], and Wen and Zhu [66]. For example, Goolsbee and Syverson [21] find that in the airline industry, incumbents drop prices to respond to the market penetration of a new player with substantial market power even before entry is confirmed, when only the possibility is evident.

This tendency of early responses is more pronounced in the IT industry, where the lifecycle of services and products is short, given the continuous development of disruptive technologies and resultant changes in consumer needs [41, 46]. Pan et al. [46] investigate how a threat of start-ups' market entry affects the R&D investments of incumbents in the IT industry, demonstrating that the latter reduce their investment plans when facing an entry threat from the former. In the context of platform economies, Wen and Zhu [66] examine the effect of a threat of platform owners' market entry on complementor behaviors, such as innovation. They find that developers reduce innovation to minimize wasteful development efforts when the threat increases but has not yet been executed.

In contrast, there has been little research directly examining how individuals or users respond ex-ante to pre-announcements. While studies such as Hoxmeier [26] investigated users' responses to pre-announcements, they primarily interpret such announcements as vendor-level signals that influence consumers' ex-post trust, rather than focusing on behavioral adjustments driven by the pre-announcement itself. In the field of Information Systems (IS), a few studies have examined the effects of pre-announcement—for instances, employees' cyberloafing behavior in response to organizational control policies [33], or investors' reactions to IT outsourcing announcements [45]—but these focus on internal stakeholders within organizational settings. In contrast, some research in other domains suggested that everyday consumers—comparable to end users in the Android ecosystem—may adjust their behavior ex-ante, such as shifting savings or spending in response to anticipated economic changes like tax policy announcements [42, 60] or inflation [38]. These studies challenge the prevailing view that demand can shift only in response to actual environmental or policy changes, indicating instead that demand could be adjusted based solely on pre-announcement. In our study's case, the pre-announcement is about the plan to enforce complementors using third-party payment systems, and the complementors only have

two options–complying with Google's plan (i.e., 30% fee) or removal from the platform. This creates a highly uncertain environment from the perspective of app users, affecting not only app prices but also service quality and even continuity. This uncertain environment could prompt preemptive changes in the demand side. Indeed, users often attribute responsibility for perceived platform instability or failures to complementors, even when such causality is ambiguous [10]. This tendency suggests that even before actual policy enforcement, users may strategically respond to announcements in ways that disadvantage small or non-compliant apps instead of abandoning the platform. In particular, prior literature on decision making under uncertainty suggest that enhanced uncertainty leads to established choices (e.g., products from more established brands) [43]. Accordingly, given that Google's pre-announcement primarily targets non-compliant third-party apps, demand could shift toward in-app purchase apps that already use Google's billing system. Moreover, in situations where market uncertainty is triggered, it is likely that these users would preemptively gravitate toward established apps with substantial traffic or opt to install apps only after a thorough review, resulting in a more established selection pattern.

In summary, our study makes the following theoretical contributions. First, we expand the literature on pre-announcement effects within platform governance. While existing research typically examines pre-announcement effects only in relation to complementors, our study analyzes the impact on the demand side. Considering the multisided nature of platforms (i.e., platform owner - complementors - users), our study offers a more nuanced understanding of pre-announcement effects. Specifically, we suggest that market dynamics after the pre-announcement are not affected only by firms' ex-ante response but also by demand-side ex-ante shifts. Additionally, we expand the literature on the impact of platform management on market dynamics. While prior studies typically focus on platform owner entry and analyze relationships between platform owners (i.e., first party) and complementors (i.e., third

parties), this study examines how changes among complementor's competitive relationships can lead to a new form of polarization, offering insights into the long-term sustainability of the Android ecosystem.

3. Research Setting

3.1. Research Context

Android apps typically adopt one or a combination of three major revenue models: in-app advertising, paid downloads, and in-app purchases. Developers can strategically integrate single or multiple revenue models according to their business strategies. Google has maintained a long-standing policy stating that all developers must use Google's billing system (i.e., Google Play billing system) when selling digital goods or content. However, this policy was not strictly enforced for apps outside the gaming category, and no concrete penalties for non-compliance were specified. Therefore, developers for non-game apps have used a third-party billing system charging cheaper commission fees (i.e., 3-4%) instead of Google's billing systems (i.e., 30%).

In early July 2020, major newspapers in South Korea published a series of articles that leaked Google's plan to implement its revised IAPP with a one-year grace period. It was the first occasion that the issue was publicized on a large scale, with public sources uncovering that Google would mandate all developers and vendors of digital content products and services to use the platform's billing system. The revised policy contains a plan to impose a hefty penalty on developers using 3rd party billing systems (e.g., removing non-compliant apps from the GPS or prohibiting their updates on these). The public disclosure of the IAPP drew significant attention from consumers, developers, and regulators in Korea due to concerns about potential price increases and a decline in service quality across the app market. This is supported by the statistics, showing that keyword searches related to 'in-app purchases' rose by approximately 44.67% on a six-month average (i.e., 11.94% on a four-week average)

following the information leak. These developments make this setting particularly suitable for investigating how the *market responded* to the policy announcement.

As debates continued to surround the potential consequences of the IAPP, a series of bills were proposed in Korea after the announcement to rein in the technology giant's abuse of its dominant market position—a government move that has drawn global attention for being "the first curb [on the technology behemoth's power] by a major economy [9]." In response, Google announced a series of minor modifications in the policy, including a reduction in commissions and an extension of the grace period on compliance, particularly for small app developers, and monitored the progress of the bills and public sentiment in Korea.

3.2. Data

We utilized large-scale panel data on Android users' mobile app usage behaviors obtained from the Korean market research firm Nielsen Korean Click. The company assembled a panel of Android users through stratified sampling based on diverse demographic factors such as gender, age, income, education level, and place of residence to ensure representation of the Korean population. App consumption data of each user in the panel was collected via the firm's tracking application. This data includes the frequency and duration (measured in seconds) of each user's app usage on a weekly basis. Specifically, we obtained weekly information on the app usage behaviors of 2,098 Android users who maintained their panel membership throughout 2019 and 2020. In addition, we crawled the Google PlayStore platform in February 2022 to gather additional information about each app, including details about its developers and revenue models, such as whether the app (1) offers in-app purchase (IAP), (2) is free or paid, and (3) includes in-app advertisements. Since there was a time gap between the event and the moment of crawling, we later checked the consistency of revenue models by comparing data measured in 2022 and confirmed that at least 98.88% of apps have

consistent revenue models (Please find 5.5 for more detail). Since game apps have long been prohibited from using third-party billing systems, even before the policy announcement, we excluded apps categorized as 'Games' or those uncategorized according to Nielsen's classification hierarchy. We also excluded apps published only for third-party app markets (e.g., One Store) and launcher apps[5] that cannot be directly controlled or deleted by users. Additionally, we excluded browser apps, including 'Google Chrome' and 'Samsung Internet Browser,' as these are used for multiple purposes, spanning various categories from games to utilities.

We focus on the number of new users to provide app developers with insights into the effects of the IAPP announcement on market response. We identified new users on the basis of whether they had not used a particular app over a given duration (i.e., the last six months). In addition, we also leveraged the number of new users who had specific demographic characteristics such as low or high education level and usage patterns such as using more or less diverse apps and using multiple app markets before the announcement. The descriptive statistics and the definitions of each variable are provided in Table 1.

[Table 1 about here]

3.3. Empirical Framework

We leverage the circumstance that the IAPP specifically targets apps published by developers selling digital content and services through third-party in-app billing systems. Also, as highlighted in Section 3.1., the pre-announcement of Google caused by the disclosure of internal information by a major newspaper was unexpected but also impactful, showing a dramatic increase in the search volume related to this issue after the information leakage (See

[5] A launcher app represents the app managing the home screen and launching apps on Android devices. Every Android phone comes with a pre-selected default launcher, automatically installed during manufacturing, so users cannot choose or control its usage.

5.3.). This setting offers us a unique opportunity to apply the quasi-experimental approach to the exploration of the IAPP announcement. Since we aimed to observe how the market responded to Google's pre-announcement, which would be executed on apps labeled as 'IAPs,' we compared Android apps that were labeled as providing 'IAPs' (the treatment group) with apps that were not labeled as providing 'IAPs' (the control group) in the Google Play Store. The first leakage of the information happened on July 9, 2020. Given that the dataset considers Monday as the first day of the week, the weeks from January 20 to June 29, 2020 (pre-announcement period) and the weeks from July 6 to December 14, 2020 (post-announcement period) were analyzed. The pre-announcement period ended on July 6, 2020, the date of the first article stating that Google would be strengthening its IAPP.

Our baseline econometric model is the following Difference-in-Difference (DID) specification at the app-week level as described in Equation (1):

$$New\ Users_{it} = \beta_0 + \beta_1 Treat_{it} + \mu_i + \delta_t + \varepsilon_{it} \quad (1)$$

$Treat_{it}$ is a binary variable representing whether the app is labeled as 'IAP' and activated in week *t* from when the IAPP was announced (i.e., from July 6 to 12, 2020). We include the two-way fixed effects, μ_i and δ_t, to control for the unobserved app- and week-specific confounders, respectively.

In addition to the main DID model, we also implemented the triple DID model (i.e., DDD) to examine the moderating effects of app characteristics such as popularity (i.e., Rank) and revenue models operating. The DDD model specification is described in Equation (2):

$$New\ Users_{it} = \beta_0 + \beta_1 Treat_{it} + \beta_2 Treat_{it} \cdot Rank_i + \beta_3 Treat_{it} \cdot Paid_i + \beta_4 Treat_{it} \cdot Ads_i + \beta_5 Treat_{it} \cdot Paid_i \cdot Ads_i + \mu_i + \delta_t + \varepsilon_{it} \quad (2)$$

In Equation (2) $Rank_i$ refers to the app's rank based on the average weekly user counts during the pre-treatment period within the categories where the app belongs; $Paid_i$ indicates whether the app costs users for downloading; Ads_i indicates whether the app has an in-app

advertisement or not. We also include app fixed effects (μ_i) and week fixed effects (δ_t) to control for unobserved heterogeneity across apps and temporal shocks, respectively.

To get insights on whether the app acquires new users with certain demographic and habitual patterns, we also replicated the main analyses with multiple dependent variables representing the number of new users but with certain characteristics. We employed *#High(MidHigh/MidLow/Low) IncNewUsers,* which refers to the number of new users whose income level is high(mid-high, mid-low, low) based on the classification by quartiles of panel population. Regarding the app usage patterns, we used *#More(Less) AppNewUsers* and *#Multi(Single)home NewUsers*, which refers to the number of new users who have a larger (less) number of used apps than average during the pre-treatment period and the number of new users who used multiple (single; Google Play Store) app markets, respectively.

4. Results

4.1. Impacts of Pre-announcement on New Users

Column (1) in Table 2 presents the results of difference-in-difference estimation for the 12 months (i.e., 48 weeks). The result shows that the pre-announcement increases the number of new users acquired for apps that had complied with Google's IAPP by about 0.7 % per week ($\beta = 0.007$, $p < 0.01$), as shown in column (1). This increase reflects users' anticipation that apps which have already adopted Google's billing systems would provide stable services with a low risk of service discontinuity or price increase, leading them to adopt IAP-compliant apps more. As a result, the pre-announcement of platform owners promoting compliance with IAPP worked as a new driver for increased new user acquisition.

To further investigate the IAPP announcement's impact on new user acquisition by app characteristics, we conducted DDD analyses with moderating variables measuring popularity and the structure of revenue models. Column (2) in Table 2 presents the results of the DDD analyses. We find that the apps that had more popularity, indicated by lower rank

values, attracted more new users (β = -1.48e-05, $p < 0.05$), as shown in column (2). This suggests that an improvement of 1,000-position in rank corresponds to approximately a 1.48 % increase in new users. For instance, an app moving from the 5,000th to the 4,000th position would experience approximately a 1.47% additional increase in new users. This is significant, considering that the number of apps enrolled in the Google Play Store is about 1.68 million apps as of September 2024, and some single categories (i.e., games, education) account for more than 11%, highlighting that rank differences within a single category can span tens of thousands. Therefore, even though the absolute value of the regression coefficient seems small, the criticality of this effect could be significant. In other words, popular apps attract more new users after the IAPP announcement, whereas unsuccessful apps experience a loss in their new user acquisitions. This result reflects users' inclination toward safer apps, driven by the perception that more established apps are less likely to be suspended or decrease their service quality.

Additionally, we observe heterogeneous effects across different revenue models–download fees, in-app advertisements, or both—on IAP apps after the pre-announcement. For IAP apps with download fees, new user acquisition declined by 1.68% (β = -0.017, $p < 0.05$ for column (2)), effectively offsetting the gains observed in other IAP apps (β = 0.016, $p < 0.05$ for column (2)). This suggests that the IAPP pre-announcement did not induce spillover effects for IAP apps with download fees, whereas other IAP apps were able to attract more new users. Given that users tend to choose IAP apps for their perceived stability and low likelihood of policy changes, it is natural that they would avoid apps requiring upfront payments before usage—especially in a highly uncertain market environment. This aligns with prior research showing that users consider factors related to revenue-model before downloading new apps [37]. In contrast, IAP apps with in-app advertisements did not show a significant difference in new user acquisition compared to apps without additional revenue

models (β = -0.002, p > 0.01 for column (2)), indicating that ad-based monetization did not influence new user acquisition following the pre-announcement. Notably, however, IAP apps operating under a multiple revenue model—combining download fees with in-app advertisements—experienced a substantial increase in new user acquisition, growing by 10.6 % (β = 0.106, p < 0.01), as shown in column (2). This finding suggests that in an uncertain environment, users may perceive multi-revenue apps as more established and financially stable, thereby reducing concerns over potential service disruptions.

[Table 2 about here]

4.2. Understanding the Composition of New Users

Along with estimating new user acquisition across all groups, we extend our analysis by examining the demographic and habitual characteristics of new users who responded to the pre-announcement. This approach allows us to better understand the composition of new users, providing valuable insights that could inform promotion or marketing strategies aimed at leveraging such events to attract a larger base of new users.

From Table 3, we find that IAP apps attracted new users across most income groups (β = 0.004, p < 0.05 for the highest income group; β = 0.004, p < 0.05 for the mid-high-income group; β = 0.003, p < 0.05 for the mid-low-income group), while they did not attract new users belonging to the low-income group. Examining usage patterns, we observe that IAP apps were more likely to acquire new users from groups that use a diverse range of apps (β = 0.007, p < 0.05 in column (1) in Table 4), compared to those with more limited app usage (β = 0.003, p < 0.10 in column (2) in Table 4). Similarly, IAP apps attracted new users from groups that frequently use multiple app markets, whereas they could not gain traction among users who primarily rely on a single app market. Individuals who actively explore various apps and engage across multiple app markets demonstrate a higher level of involvement and awareness of the app market ecosystem, making them more likely to respond proactively to

policy change pre-announcements. Conversely, those who use fewer apps and remain within a single app market exhibit lower engagement and awareness, making them less likely to adjust their behavior in anticipation of such changes.

[Table 3 and Table 4 about here]

This pattern may contribute to increasing polarization within the mobile app markets. As observed in Section 4.2, popular apps with diversified revenue streams acquired a greater number of new users following the pre-announcement (i.e., the quantitative aspect of user acquisition). However, these newly acquired users tended to have higher purchasing power or a stronger engagement with the app market, suggesting that the qualitative aspect of new users also plays a role in deepening the divide between established and non-established apps.

Alongside replicating the main models with different dependent variables, Tables 5 and 6 illustrate how app characteristics moderate user acquisition effects across different user groups. Specifically, Table 5 shows that higher-ranked apps (i.e., popular apps) experienced a more significant increase in new users with relatively higher income levels (*Treat * Rank*). Meanwhile, Table 6 indicates that having multiple revenue streams significantly impacted new user acquisition across all user segments (*Treat * Paid * Ads*).

[Table 5 and Table 6 about here]

4.3. Analysis After Implementation

Finally, to evaluate the actual impacts of the policy beyond the immediate market responses triggered by the pre-announcement, we analyzed changes in new user acquisition around the final enforcement deadline on April 1, 2022, when Google warned that non-compliant developers would be blocked from updating their apps. This analysis also serves to validate our pre-announcement findings by examining whether the observed trends persisted once enforcement took effect. Due to the limited availability of data at the time of implementation, we utilized the full available data (i.e., first half of 2021 and February 2022 onwards) to

aggregate the number of new users for each app.[6] The results from Table 7 indicate that similar to the pre-announcement period, the number of new users increased for apps explicitly marked as IAP apps. This observation is consistent with the pre-announcement findings, suggesting that users are opting for the safer choice of apps, which already indicates the use of 'in-app purchase' over those with uncertain update futures.

[Table 7 about here]

5. Robustness Checks

5.1. Parallel Trend Assumption

To employ DID as our model, we need to check whether it satisfies the parallel trend assumption between the treatment and control groups. To compare the pre-trend of the new users before pre-announcement, we conducted a relative time model (i.e., leads-and-lags models). As shown in Figure 1, we find almost no significant differences between the IAP apps and non-IAP apps before the pre-announcement for 6 months, while the IAP increases the number of new users during the whole period of the post-treatment period, 6 months. From the result, we find there is a low possibility of violating the parallel trend assumption.

5.2. Sensitivity Check for Research Window

To ensure that our main findings are not driven by the specific selection of the research period, we conducted a series of sensitivity analyses by systematically adjusting the research window. Specifically, we re-estimated our main models using pre- and post-treatment periods of three, four, five, and six months to examine the robustness of our results. As shown in Table 8, the estimated treatment effects remain consistently positive and statistically significant across all specifications, with IAP apps experiencing an increase in new user

[6] To maximize the utilization of our available data range, we defined a new user as a panel *i* who had no usage record for app *j* during the first halt of 2021 and also throughout February 2022. If panel I, who had no prior usage record for app *j*, exhibit usage of app *j* in month *t*, this instance was counted as a new user acquisition.

acquisition ranging from 2.1% to 2.9%, depending on the selected research window. These findings suggest that our results are robust and not sensitive to minor variations in the research period, reinforcing the validity of our conclusions.

5.3. Validation of the Shock Timing

The first leakage of Google's enforcement plan on non-compliers of IAPP occurred on July 9, 2020, in South Korea. We define July 9, 2020—the date of the first leakage—as the start of the post-treatment period. However, there could be a concern that the information was not prevalent among mobile users, so the effect was not caused by the pre-announcement. To address this concern, we analyzed how search volume regarding the issues on IAPP increased after the first leakage by using keyword volume data provided by a major search engine platform in South Korea. In Figure 2, the x-axis and y-axis present the week and relative search volume. The search volume range is from zero to one hundred, a standardized index that fixes the highest searched volume as one hundred and assigns the value of the remaining days relative to the highest search volume. In the graph, the vertical line located in the middle of the line graph presents when the first leakage happened. As shown in Figure 2, we find that the search volume increases immediately after the first leak, supporting the validity of our post-treatment starting point.

[Figure 1 and Figure 2 about here]

5.4. Addressing the Discrepancy in Data Collection

Although we utilized structured and comprehensive panel data covering individuals' app usage throughout the research period, we collected the app characteristic data by crawling from Google PlayStore (GPS) after the pre-announcement but prior to the enforcement. This could hinder estimating the exact impacts of IAPP pre-announcements and the moderating effects of the revenue models if each app's revenue model changes frequently or is highly volatile. We, therefore, needed to understand how frequently apps change their revenue

models over time. Fortunately, we were able to compare our current data with historical data crawled from GPS in 2013 with data from GPS in 2022, which is our current crawled data.

Table 9 shows that only 0.43% of all apps changed their revenue models during almost 10 years (i.e., from 2013 to 2022), while 99.57% retained their original revenue models. Furthermore, we also compared the revenue models of all apps between February 2022 and September 2022 (i.e., The last crawling moment). Table 10 shows that almost all apps show consistency in their revenue models, although the enforcement had already begun in April 2022. This could be another evidence that apps do not easily change their revenue models, implying that our crawled information may have a high possibility of being consistent with the reality that happened before.

[Table 8, Table 9, and Table 10 about here]

6. Discussion and Implications

Google is indisputably the dominant app powerhouse, taking the reins of operating millions of mobile apps that affect the fabric of our lives—a portfolio that extends from social networking, commercial transactions, and information search to work productivity and entertainment. As a platform owner, the Android-based titan affords developers around the world endless access to commercial and technical resources, facilitating app-based exchanges characterized by efficient matching and low transaction costs. At the same time, however, it can take advantage of the economics of platforms and their market dominance, which often compromises the interests of platform complementors (i.e., developers) and users [32]. Many experts view the enforcement of the in-app purchase requirement as an exploitation of platform monopoly and market supremacy, as Google is gearing up to impose a toll on the mobile highways through which apps are downloaded, and business is transacted. Concerns arise that this move is likely to foment inequality among developers, suppress the survivals of small developers, and affect the welfare of consumers. Our research offers a variety of initial

insights into these speculations, providing managerial and scholarly implications as to what needs to be done on legal and commercial grounds to protect "platform minorities" (i.e., new developers and users with no bargaining power), and accordingly preserve the healthy ecosystem in platforms that benefit both ends.

Our findings indicate that the intervention of the in-app purchase system could exert the most detrimental effects on complementors unfavorable to Google's strengthened control and relatively unestablished apps, widening the gap between complementors with respect to operational revenues and future growth opportunities. Building on the existing research on supplier participation in digital platforms [29] suggesting that platform owners' intervention can influence participation thresholds of complementors, our findings suggest that the interventions by platform owners can even exacerbate competitive asymmetries, reinforcing advantages of already dominant complementors. One of the primary purposes that drove Google to initially announce plans to strengthen its payment policy was to regulate not small-scale developers but established giants, which had avoided in-app purchase fees by capitalizing on their customers' loyalty and inducing them to transact payments outside their apps [65]. However, counter to expectations, established developers that enjoy scale, breadth, and resource abundance (e.g., domestic SNS platforms with enormous user bases) have benefited from a sizable increase in the number of active users following the announcement of the new payment policy. Given the ongoing uncertainty and dynamic evolution of the app market due primarily to the threat of mandatory payment, consumers tend to switch to free apps or apps with no in-app purchase features, most of which are plied by large developers. Incumbent players that focus on a single category or offer numerous apps falling under multiple categories can capitalize on their own network externalities with which to generate revenue from both free (i.e., advertisements) and paid (i.e., in-app purchases) apps [2]. Conversely, many small developers who are unequipped with the necessary financial

resources and technical expertise cannot leverage positive network effects, further losing customers as competition with their more established rivals intensifies. Google's payment intervention will compel existing small developers to raise the prices of their app services and bring pressure to bear on them to substantially increase the quality of their digital products—all while reinforcing barriers to the entry of start-up developers into the platform. This unleveled playing field will eventually expand the chasm between the small and the incumbent, driving many of the former from the market.

The mandatory payment, upon its full implementation, could open new windows of opportunities for incumbent developers to broaden and consolidate the market in the long run as competition against them weakens, albeit initial payments to the platform owner are due in the short run. Recently, Google accorded some established developers "special treatment," which enabled them to continue servicing their customers through external payment options juxtapositioned with the Google billing system. For example, Google and Spotify have recently entered into an agreement to implement expanded payment options, through which the popular streaming service provider is empowered to use its own payment system alongside GPS's billing scheme [58]. This strategic partnership between platform owners and established developers is expected to further penalize small players who might have to bear the twofold blow of platform regulation and intervention, as many of them operate under the influence of their larger counterparts (i.e., SNS portals). We advise that expanded payment first be granted to small developers and then progressively applied to all other participants in the market. Any alliances and interactions between platform owners and incumbent developers should also be more transparent and pro-competition so that such corporations enhance value-creation and innovation for all platform participants, not merely for themselves.

Google's in-app purchase requirement may damage the spirit with which open

platforms were created [47], but what could be even more concerning is applying a standard commission structure across developers, irrespective of size and revenue. We urge regulators and Google to loosen Android payment restrictions on small developers by substantially cutting in-app transaction fees—a measure that offers small players leeway in cultivating their technical and financial capabilities. This adjustment will likely work in Google's favor because the participation of more developers translates to increased opportunities for the platform to succeed and excel [48]. Moreover, because developer engagement is crucial for the expansion and value-creation of platforms [63], offering some latitude and protection to small players is advantageous given that these participants constantly infuse an entrepreneurship spirit into the platform's ecosystem, apart from continually supplying content to it.

Also, although it is premature to predict how and to what extent developers will increase the prices of their offerings when Google's payment mandate is strictly enforced, our findings are a preview of consumer reactions to price increases that will arise from the mandate. Many large developers, particularly local over-the-top (OTT) service providers in Korea, increased their subscription fees for streaming services even before the actual implementation of the payment regulation. Amid the threat of policy intervention, developers of various content products (i.e., movies, music, webtoons) are on the cusp of following suit but have temporarily stayed a response given the apparent negative pushback and complaints from consumers and regulators. While the reaction of incumbent developers is easily predictable, it is less obvious how their small, independent equivalents will respond and adjust prices. We recommend that Google elevate its relationships with small developers, viewing them as strategic partners for value co-creation [28, 40] and thereby maintain the openness of its ecosystem and spur innovation—a founding strategic vision and spirit of the Google platform [1]. Under this orientation, the technology giant should offer small

developers not only a reduced commission structure but also full technical support for integrating the billing system into their networks, which also financially burdens the latter. Platform operators should be mindful that the design and deployment of platform governance rules have a profound impact on complementors' actions and survival, which, in turn, affect the attractiveness, value, and survival of platforms themselves [34, 50].

Our study likewise presents a number of scholarly implications. Researchers have long scrutinized the economic and social consequences of monopoly in digital platforms, which are, by design, two-sided [51]. The relevance of platform monopoly to consumer welfare has been intensively explored, documented, and understood (e.g., [14]), but most of these works have been carried out on the basis of legal reports, business anecdotes, or game-theoretic analytical derivations (e.g., [55, 71]). Although useful, these approaches are conceptual in principle, uncovering limited empirical insights that reflect the dynamics of a platform and the behaviors of its participants. Moreover, empirical research that revolves around the effects of platform operators' interventions on the supply side of the platform (i.e., developers) remains scarce. Two exceptions are the works of Foerderer et al. [17] and Wen and Zhu [66], which recently explored how platform owners' direct entry into complementors' markets affects complementary innovation in the ecosystem and pricing and value-creation strategies, respectively.

The threat of implementing Google's payment option requirement is truly a paradigm-shifting policy intervention that massively affects both the demand and supply sides of the platform, as well as its governance, coordination, competition, and innovation. Because previous studies investigated each side of a platform in isolation, limited progress has been made in the holistic understanding of the interaction between the two sides of such a platform. Assessing the two sides of a platform concurrently provides us a more comprehensive grasp of the dynamics of platform ecosystems in the event of major policy

changes. Similarly, research on platform governance has focused exclusively on tension and collaboration only between platform owners and developers [17, 21]. As our results show, the major policy intervention exerts heterogeneous effects, depending on the size of the platform developers. Previous scholarly works related to platform governance disregarded such diverse effects, which vary depending on developer characteristics. Our study attempted to fill these voids by concentrating on dynamics of policy effects and factoring variations in developers into the analysis. Under certain circumstances, local incumbent developers could be as influential as global platform owners, with the former wielding enormous bargaining power against the latter. In platform ecosystems, cooperation can occur in the shadow of competition. That is, these two parties appear to be in dispute at a glance, but they implicitly establish a form of strategic partnership in pursuit of mutual benefits, which adversely affects small developers. Platform economics is complex given that drawing a distinct line between competition and cooperation is elusive, and power imbalance varies and evolves situationally within and across platform participants.

Finally, this research expands the literature on market signaling via pre-announcements of strategic actions [24]. Studies in the fields of strategy and finance often regard announcements or previews of a potential action as a signal and not the action itself [24, p.404]. Put differently, pre-announcements serve as market signals of senders' impending actions, which stimulate strategic responses from signal receivers. As for digital platforms, owners can deliberately or unintentionally send signals to strengthen their control and govern the behaviors of complementors and consumers. Similar to operations in stock markets and competitive corporate environments, signals of actions can often be even more impactful than actual actions because the former more actively determines public perceptions and reactions early in the process than the latter [66].

Companies ought to react to market signals rapidly from the onset of the process and

before an actual event takes place to maximize competitive gains early. App developers, in particular, exhibit *ex-ante* strategic reactions upon receiving the signals sent by app stores, as doing so can preempt competition against rivals and maximize possible rewards [66]. Likewise, app consumers are forward-looking agents who rationally adjust their consumption in accordance with their current and future demands in an effort to maximize their consumption utility [35]. These consumers also have a reason to behave strategically because many in-app prices are hidden in the sense that they are discoverable only in the final payment process but are unavailable before or during searches [55]. The announcement of the mandatory in-app purchase system strongly communicates Google's strategic vision and governance for the future to app developers and consumers [27]. These signals will, therefore, likely drive developers and consumers to exhibit strategic behaviors in turn for maximized utility and value before actual enforcement. This is consistent with Soh and Grover [57], who conceptualize platform boundary resources—including announcements and policy changes—not merely as technical tools, but as interpretive resources that enable distributed sensemaking across ecosystem participants. From this perspective, Google's policy announcement act as a boundary resource that catalyzes a shared interpretive process among users and developers, leading to anticipatory strategic behaviors. Our findings shed light on how platform signals from owners in the form of pre-announcements of mandatory rulings initiate strategic reactions from platform stakeholders, including developers, consumers, and policymakers. Platform operators wait for market responses following pre-announcement and then determine the appropriate timing and level of actual implementation accordingly. The antitrust implications we identified in the context of platform economics and market signaling can motivate researchers to assess how and why platform signals can have differential effects and consequences on platform participants and what this means for the health of digital platform ecosystems amid major policy interventions. While these shifts

may initially disadvantage smaller or less-resourced developers, they also present a potential window of strategic opportunity. Prior research shows that timely responses to platform-level innovations can enhance app performance [56]. Extending this logic, we propose that smaller developers could improve their competitiveness by proactively interpreting such announcements as early market signals and rapidly adapting their app offerings or communication strategies accordingly. In this sense, policy announcements—often perceived as threats—could be reframed as actionable events that small developers leverage to reposition themselves and capture new demand segments.

7. Conclusion

The mobile app market is going through a period of disequilibrium, as Google's restrictions on Android's in-app purchases have unsettled the complexity and dynamics of the platform and its ecosystem. Our results suggested that the platform policy intervention may exacerbate inequality among complementors that vary widely in size and capacity. Established developers can benefit from policy enforcement but smaller developers, who operate under the dual pressures of platform owners and established competitors, are unlikely to gain similar advantages. On the basis of comprehensive mobile app data, we derived a variety of initial insights into how the in-app purchase regulation would affect the fabric of the ecosystem and change the nature and dynamics of competition via consumption changes. We also clarified what these forthcoming changes mean to platform governance and the welfare of its stakeholders.

Our study is not without limitations. Our empirical results are based on the analyses of mobile app data collected in one country (i.e., South Korea), where Google's Android platform enjoys a commanding share of the market for mobile operating systems. We, therefore, made no attempt to generalize our findings to other nations. Yet, the Android mobile OS outrivals that of Apple in many parts of the world, and the former has successfully

maintained a global market share in the neighborhood of 70%—a figure that closely matches its dominance in Korea. Another caveat is that our findings reflect the reactions of consumers and developers that occurred only at the initial stage of the policy announcement. Since the policy is still under development, an ongoing investigation over extended periods would advance a more comprehensive understanding of the effects of the intervention. Finally, at the time of this study, Google faced numerous legal challenges due to possible antitrust violations in many jurisdictions. It is important to understand how the company's manipulation of mobile payment structures and the subsequent extraction of additional rents from the practice are legally evaluated. Amid this backdrop, we did not fully take legal developments into account in our analyses but tackled issues mainly from economics and strategic perspectives that surround platform governance. Given the significance of legality in assessing the effects of platform monopoly, caution should be exercised when interpreting our results and suggestions. We invite other researchers to fill these voids and explore other important issues revolving around the in-app purchase restriction.

Tables and Figures

Table 1. Descriptive Statistics

	Variables	Definition	Obs	Mean	SD	Min	Max
Dependent Variables	*# New Users*	The number of new users for app *i* at week *t*	254,736	0.19	0.91	0	199
	# New Users with Low Edu	The number of new users who have not attended college for app *i* at week *t*	254,736	0.05	0.30	0	46
	# New Users with High Edu	The number of new users who have attended college for app *i* at week *t*	254,736	0.14	0.70	0	153
	# New Users with More App Usage	The number of new users whose average number of app used in the pre-treatment period is above average for app *i* at week *t*	254,736	0.13	0.65	0	147
	# New Users with Less App Usage	The number of new users whose average number of app used in the pre-treatment period is above average for app *i* at week *t*	254,736	0.06	0.38	0	52
	# New Users with Multi-homing	The number of new users who used diverse app markets in the pre-treatment period for app *i* at week *t*	254,736	0.17	0.82	0	181
	# New Users with Single-homing	The number of new users who only used GPS as an app market in the pre-treatment period for app *i* at week *t*	254,736	0.02	0.17	0	18
Treatment	*In-App*	Whether the app *i* is an in-app purchase app or not	254,736	0.22	0.41	0	1
Covariates	*Down*	Whether the app *i* is a paid app (paid before download)	254,736	0.02	0.13	0	1
	Ads	Whether the app *i* has in-app advertisement	254,736	0.46	0.50	0	1
	App Rank	Rank of app *i* within the same category, based on the total number of users during the pre-treatment period	254,736	468.98	414.56	1	1529

Table 2. Impact of IAPP Pre-announcement

Variables	*#NewUsers*	*#NewUsers*
	(1)	(2)
Treat	0.007***	0.016**
	(0.003)	(0.008)
*Treat * Rank*		-1.48e-05**
		(6.59e-06)
*Treat * Paid*		-0.017**
		(0.00748)
*Treat * Ads*		-0.002
		(0.007)
*Treat * Paid * Ads*		0.192***
		(0.024)
Constant	0.106***	0.106***
	(0.003)	(0.003)
Number of Applications	5,307	5,307

Observations	254,736	254,736
R-squared	0.002	0.003

Note(s). Standard errors in parentheses are clustered by apps. Two-way fixed effects (App, Week) are controlled. *** p<0.01, ** p<0.05, * p<0.1.

Table 3. Impact on New Users from Each Income Level Group

Variables	*Household Income*			
	#HighInc NewUsers	***#MidHighInc NewUsers***	***#MidLowInc NewUsers***	***#LowInc NewUsers***
	(1)	**(2)**	**(3)**	**(4)**
Treat	0.004**	0.004**	0.003**	0.001
	(0.002)	(0.002)	(0.001)	(0.000)
Constant	0.048***	0.050***	0.025***	0.003***
	(0.000)	(0.002)	(0.002)	(0.001)
Number of Apps	5,307	5,307	5,307	5,307
Observations	254,736	254,736	254,736	254,736
R-squared	0.001	0.001	0.001	0.000

Note(s). Standard errors in parentheses are clustered by apps. Two-way fixed effects (App, Week) are controlled. *** p<0.01, ** p<0.05, * p<0.1.

Table 4. Impact on New Users from Each App Usage Group

Variables	*App Consumption*		*App Market Usage*	
	# MoreApp NewUsers	***# LessApp NewUsers***	***# Multihome NewUsers***	***# Singlehome NewUsers***
	(1)	**(2)**	**(3)**	**(4)**
Treat	0.007**	0.003*	0.008**	-0.001
	(0.002)	(0.002)	(0.003)	(0.001)
Constant	0.080***	0.037***	0.097***	0.013***
	(0.003)	(0.002)	(0.003)	(0.001)
Number of Apps	5,307	5,307	5,307	5,307
Observations	254,736	254,736	254,736	254,736
R-squared	0.002	0.001	0.002	0.001

Note(s). Standard errors in parentheses are clustered by apps. Two-way fixed effects (App, Week) are controlled. *** p<0.01, ** p<0.05, * p<0.1.

Table 5. Heterogeneous Impacts on New Users from each Income Level Group

Variables	**Household Income**			
	#HighInc NewUsers	***#MidHighInc NewUsers***	***#MidLowInc NewUsers***	***#LowInc NewUsers***
	(1)	**(2)**	**(3)**	**(4)**
Treat	0.009*	0.011*	0.007**	4.11e-04
	(0.005)	(0.006)	(0.003)	(0.001)
*Treat * Rank*	-8.06e-06*	-1.11e-05**	-6.33e-06**	-1.50e-06
	(4.08e-06)	(4.58e-06)	(2.68e-06)	(9.35e-07)
*Treat * Paid*	-0.010**	-0.007	-0.004	-4.18e-04
	(0.005)	(0.005)	(0.003)	(9.56e-04)
*Treat * Ads*	3.60e-04	-0.004	-0.001	0.002*
	(0.004)	(0.005)	(0.003)	(0.001)
*Treat * Paid * Ads*	0.133***	0.087***	0.073***	0.027***
	(0.024)	(0.006)	(0.003)	(0.001)
Constant	0.048***	0.042***	0.019***	0.003***

	(0.002)	(0.001)	(0.001)	(0.001)
Number of Apps	5,307	5,307	5,307	5,307
Observations	254,736	254,736	254,736	254,736
R-squared	0.001	0.001	0.000	0.000

Note(s). Standard errors in parentheses are clustered by apps. Two-way fixed effects (App, Week) are controlled. *** p<0.01, * * p<0.05, * p<0.1.

Table 6. Heterogeneous Impact on New Users from Each App Usage Group

Variables	***App Consumption***		***App Market Usage***	
	# MoreApp NewUsers	***# LessApp NewUsers***	***# Multihome NewUsers***	***# Singlehome NewUsers***
	(1)	**(2)**	**(3)**	**(4)**
Treat	0.013*	0.008	0.017**	3.66e-04
	(0.007)	(0.005)	(0.008)	(0.003)
*Treat * Rank*	-1.58e-05**	-5.36e-06	-1.73e-05***	1.48e-06
	(5.64e-06)	(3.98e-06)	(6.33e-06)	(2.03e-06)
*Treat * Paid*	-0.014**	-0.005	-0.017**	-8.21e-04
	(0.007)	(0.004)	(0.007)	(0.002)
*Treat * Ads*	5.87e-04	-0.003	-5.52e-06	-0.002
	(0.006)	(0.004)	(0.007)	(0.002)
*Treat * Paid * Ads*	0.178***	0.029***	0.182***	0.048***
	(0.024)	(0.004)	(0.024)	(0.002)
Constant	0.062***	0.037***	0.080***	0.014***
	(0.001)	(0.002)	(0.001)	(4.03e-04)
Number of Apps	5,307	5,307	5,307	5,307
Observations	254,736	254,736	254,736	254,736
R-squared	0.001	0.000	0.001	0.000

Note(s). Standard errors in parentheses are clustered by apps. Two-way fixed effects (App, Week) are controlled. *** p<0.01, * * p<0.05, * p<0.1.

Table 7. Impact of IAPP Implementation on New Users

Variables	***#NewUsers***	***#NewUsers***
	(1)	**(2)**
Treat	0.014**	0.016**
	(0.005)	(0.008)
*Treat * Rank*		-1.48e-05**
		(6.59e-06)
*Treat * Paid*		-0.017**
		(0.00748)
*Treat * Ads*		-0.002
		(0.007)
*Treat * Paid * Ads*		0.192***
		(0.024)
Constant	0.223***	0.106***
	(0.003)	(0.003)
Number of Applications	8,425	5,307
Observations	84,250	254,736
R-squared	0.005	0.003

Note(s). Standard errors in parentheses are clustered by apps. Two-way fixed effects (App, Week) are controlled. Due to the li mitation of the data collection period, it has four weeks of pre-treatment period and six weeks of post-treatment period. *** p< 0.01, ** p<0.05, * p<0.1.

Figure 1. Result from Relative Time Model

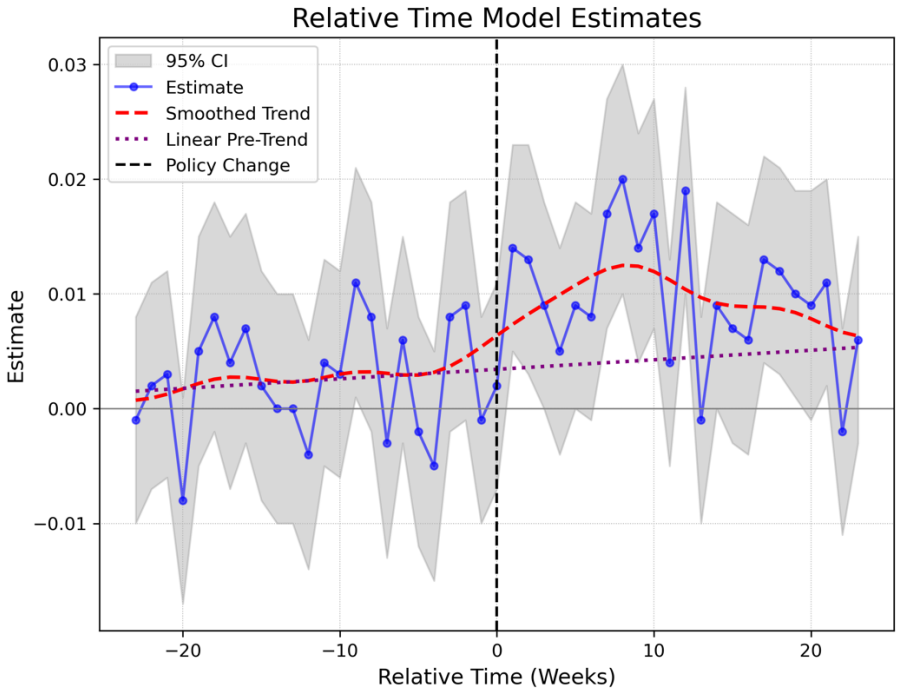


Table 8. Sensitivity Check for Research Window

Variables	***#NewUsers***			
	(1)	**(2)**	**(3)**	**(4)**
	3 months	**4 months**	**5 months**	**6 months**
Treat	0.024**	0.021**	0.027***	0.029***
	(0.008)	(0.008)	(0.009)	(0.009)
Constant	0.216***	0.208***	0.210***	0.212***
	(0.008)	(0.008)	(0.008)	(0.010)
Number of Applications	5,307	5,307	5,307	5,307
Observations	127,368	169,824	212,280	254,736
R-squared	0.001	0.002	0.001	0.001

Note(s). Standard errors in parentheses are clustered by apps. Two-way fixed effects (App, Week) are controlled. Due to the limitation of the data collection period, it has four weeks of pre-treatment period and six weeks of post-treatment period. *** p<0.01, ** p<0.05, * p<0.1.

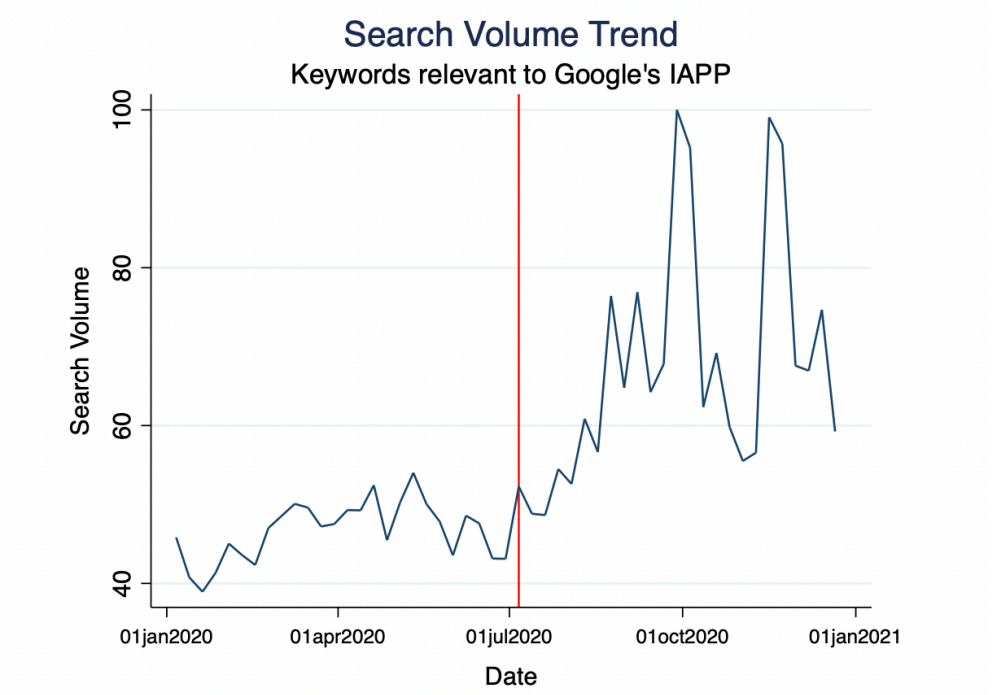
Table 9. Revenue Model Comparison Between 2013 and 2022

2013 vs. 2022	***Charging Download***	
	Freq.	**Percent (%)**
Inconsistent	6	0.43
Consistent	1,375	99.57

Table 10. Revenue Model Comparison between February and September 2022

Comparison	*Charging Download*		*In-App Purchase*		*In-App Advertisement*	
	Freq.	**Percent (%)**	**Freq.**	**Percent (%)**	**Freq.**	**Percent (%)**
Inconsistent	6	0.07	96	1.12	54	0.63
Consistent	8,549	99.93	8,459	98.88	8,501	99.37

Figure 2. Search Volume Trend on Google' IAPP



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